Paleomagnetism of tills and associated paleosols in southwestern Alberta and northern Montana: evidence for Late Pliocene – Early Pleistocene glaciations[1]

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Abstract: Sequences of pre-Wisconsinan till and intercalated paleosols were sampled for paleomagnetic study. The tills were deposited during successive glaciations and the paleosols formed during interglacial intervals. Paleoargillic horizons of the paleosols and the carbonate cement (calcrete) found in some till–paleosol units generally yielded excellent data. Magnetizations of paleosols probably were acquired during the formation of the paleosols rather than during initial deposition of the tills in which they were developed. At Mokowan Butte (Alberta), the lowest paleosol has normal polarity, two of the middle tills have reversed polarity, and the uppermost till – paleosol unit has normal polarity. At Saint Mary Ridge (Montana), three of the lower tills have reversed polarity, and the upper two till–paleosol units have normal polarity. At Two Medicine Ridge (Montana), the lowest three tills are reversed, but the paleoargillic horizon on the uppermost (fourth) till is normal. Magnetostratigraphic correlation indicates that at least six glacial and six interglacial episodes are represented in the Kennedy Drift. The upper normal polarity units are interpreted as having been developed during the Brunhes Normal Chron, the underlying reversed polarity sediments during the Matuyama Reversed Chron, and the lowest normal polarity unit at Mokowan Butte during the Gauss Normal Chron. The oldest glaciations here extend into the Pliocene (2600 ka), making these sediments among the oldest glacial deposits in North America. Alternatively, the lowest normally magnetized paleosol at Mokowan Butte may have formed during either the Jaramillo or the Olduvai subchrons, although this is considered less likely.

Résumé : Des séquences de tills anté-wisconsiniens et de paléosols intercalés ont été échantillonnées pour une étude paléomagnétique. Les tills se sont déposés durant les épisodes de glaciations successives et les paléosols se sont développés durant les intervalles interglaciaires. Les horizons paléoargiliques des paléosols et le ciment de carbonate (encroûtement calcaire) trouvé dans certaines unités de till–paléosol ont fourni, en général, des donnés très satisfaisantes. Les aimantations des paléosols furent acquises probablement durant la période de développement des paléosols, plutôt que durant l'épisode de dépôt initial des tills sur lesquels ils se sont formés. À Mokowan Butte (Alberta), le paléosol de la base présente une polarité normale, deux des tills médians ont une polarité inverse et l'unité de till–paléosol sommitale affiche une polarité normale. À Saint Mary Ridge (Montana), trois des tills inférieurs ont une polarité inverse, et deux des unités de till–paléosol sommitales possèdent une polarité normale. À Two Medicine Ridge (Montana), les trois tills inférieurs sont à polarité inverse, mais leur horizon paléoargilique développé sur le till sommital (quatrième) est à polarité normale. Une mise en corrélation magnétostratigraphique révèle l'existence d'au moins six épisodes glaciaires et six interglaciaires dans le Drift de Kennedy. Les unités sommitales de polarité normale sont interprétées comme ayant été déposées durant le Chron normal de Brunhes, les sédiments sous-jacents de polarité inverse ayant été déposés durant le Chron inverse de Matuyama, et l'unité inférieure de polarité normale à Mokowan

Received June 27, 1994. Accepted December 9, 1994.

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Butte ayant été formée durant le Chron normal de Gauss. Ici, les plus anciennes glaciations reculent jusqu'au Pliocène (2600 ka), associant leurs sédiments aux plus anciens dépôts glaciaires de l'Amérique du Nord. D'autre part, le paléosol de la base à Mokowan Butte, d'aimantation normale, a pu être formé durant l'un ou l'autre des subchrons de Jaramillo ou d'Olduvai, bien qu'une telle interprétation soit moins plausible.
[Traduit par la rédaction]

Introduction

In this paper, the paleomagnetism of stacked sequences of pre-Wisconsinan tills and intercalated paleosols (the Kennedy Drift of Horberg (1956)) is described and used to improve estimates of age and of correlations among sequences. Earlier workers have suggested that the age of the drift is either predominantly Middle Pleistocene (Richmond 1986) or, alternatively, spans the interval between Late Pliocene and Middle Pleistocene (Karlstrom 1981, 1991). Lack of suitable materials prevents the application of radiometric dating. Thus, we have tested these ideas using paleomagnetic methods.

The till–paleosol sequences occur on isolated remnants of the Miocene–Pliocene Flaxville surface in northern Montana and southwestern Alberta, along the eastern margin of Waterton–Glacier Park (Fig. 1). Horberg (1956) considered the Kennedy Drift at Two Medicine Ridge to be of Kansan age (220–440 ka). Richmond (1965) subdivided the drift into three tills, which he correlated with the Illinoian (120–170 ka), Kansan, and Nebraskan (600–860 ka) stages. Later, Richmond (1986) reduced his estimate of the age of the Kennedy Drift to 428–780 ka. In contrast, Karlstrom (1987, 1991) gave much older age estimates on the basis of the relative degree of soil development (Harden (1982) profile and clay accumulation indices), paleomagnetic data from eight samples from Mokowan Butte, a uranium-trend date of 440 ± 120 ka from Saint Mary Ridge (Richmond 1986), and correlation with dated paleoclimatic sequences elsewhere. He argued that 10^5–10^6 years were required to form the more well-developed paleosols, and 10^4–10^5 years to form the less well-developed paleosols. On this basis, he calculated that the lowermost till at Mokowan Butte is >2600 ka.

Paleomagnetism should discriminate between these very different proposals. If the tills and paleosols are entirely younger than 800 ka, as Horberg (1956) and Richmond (1986) suggested, normal polarity should predominate, and reversed polarity should occur only at the base. Alternatively, if they range in age down to 2200 ka, as Karlstrom (1987, 1991) suggested, then the tills and paleosols should be predominantly reversed, with normal polarity only at the top of the sequences.

Geologic setting

Several erosion surfaces of different ages occur east of the Lewis Range of the Rocky Mountains (Alden 1924). The most prominent of these is the Miocene–Pliocene Flaxville Bench, upon which the Kennedy Drift rests (Alden 1924). Jungerius (1966, 1967) and Karlstrom (1987, 1990) considered this, and other similar surfaces, to be pediments formed mainly by fluvial or glaciofluvial erosion. In our study area, the Flaxville surface bevels Upper Cretaceous sandstone and shale. Flaxville remnants today form interfluves between modern valleys, on which sequences of Kennedy Drift, 30–70 m thick, are exposed (Alden and Stebinger 1913; Horberg 1956). Bedrock is exposed at Mokowan Butte and Two Medicine Ridge but not at Saint Mary Ridge.

For the purposes of this study, a unit is defined as a diamict and the paleoargillic horizons (paleosols) developed in it. Diamicts are interpreted to be glacial tills, and each diamict–paleosol unit is considered to record at least one glacial and one interglacial interval.

Tills of the Kennedy Drift are poorly sorted, consisting of about 50% Precambrian Belt – Purcell clasts in a sandy loam matrix. Where unweathered, the rock fragments are 65–78% greenish and reddish argillite, 20–32% limestone and dolomite, 7–8% quartzite, 1–4% sandstone, and 1–4% diorite and basalt (Karlstrom 1987). Calcrete, or cemented diamict, occurs as distinct layers that have formed mostly by the leaching and reprecipitation of carbonate at depth, and in some cases, by the lateral movement of groundwater (Karlstrom 1987, 1988). Calcrete horizons are not present in every unit, and their thickness varies.

A glacial origin for the tills is indicated by the abundance of striations on argillite clasts, and by their geomorphic association with, and similarities to, tills in nearby younger morainic deposits (Alden and Stebinger 1913; Karlstrom 1987). In addition, scanning electron microscope analysis indicates that about half of 35 quartz grains studied from the upper four units at Mokowan Butte have glacial textures.

Paleosols include yellowish-red, clay-rich, leached Bt or Bw horizons overlying leached Bw and (or) Bk or Bkm (calcrete) horizons (Karlstrom 1987, 1988; White 1979). They are considered to have formed mainly under interglacial conditions warmer and wetter than those of the present (Karlstrom 1987, 1988, 1991).

Karlstrom (1981, 1988) suggested that there are as many as five paleosols in five tills at Mokowan Butte and Saint Mary Ridge, and three paleosols in four tills at Two Medicine Ridge. Equating tills with glacial conditions, he argued that at least five glacial and five interglacial intervals are recorded by the Kennedy Drift. The base of the section at Saint Mary Ridge is not exposed (talus covered), so more than five tills could be present there. Geomorphic relations and relative degree of soil development indicate that Kennedy Drift predates the Wisconsinan-aged tills that are present in the modern stream valleys.

Procedures

Mokowan Butte, in southwestern Alberta, has the most complete and best preserved sequence of tills and paleosols, and has previously been studied paleomagnetically (eight samples) by Karlstrom (1987) and later by Barendregt et al. (1991). The latter workers were unable to obtain meaningful data from the lower units, which we have therefore resampled. Results from good, but less complete, sequences at

Saint Mary Ridge and Two Medicine Ridge in Montana are also reported. These localities are of special interest because they contain cemented diamict or calcrete, which provided specimens that could be thermally demagnetized, allowing the carrier of magnetization to be determined.

Vertical faces were cut in paleosols and unconsolidated fine-grained diamict, and cylindrical plastic holders 2.54 cm in diameter were inserted into the sediment. Altogether, 121 specimens were obtained, 51 from six sites at Two Medicine Ridge, 34 from four sites at Saint Mary Ridge, and 36 from four sites on Mokowan Butte. Cores of calcrete were taken with a gasoline-powered drill; 16 were collected from Two Medicine Ridge and 41 from Saint Mary Ridge, and 82 cylindrical specimens were cut from them. Including the previous sampling by Barendregt et al. (1991) at Mokowan Butte (125 specimens), a total of 331 specimens have now been collected from the Kennedy Drift: 164 from Mokowan Butte, 85 from Saint Mary Ridge, and 81 from Two Medicine Ridge.

Measurements were made on automated Schonstedt spinner magnetometers. For alternating-field (AF) demagnetization, a Schonstedt SSM-1 was used for fields up to 100 mT, and a Sapphire Instrument SI-4 for fields between 100 and 180 mT. A Schonstedt TSD-1 was used for thermal demagnetization. Most specimens were demagnetized using at least three steps, and directions were determined by line fitting. At a few sites, where pilot specimens showed unidirectional magnetization, the remaining specimens were cleaned at a single level (usually 70 mT), which was adequate to determine polarity.

Results

Stratigraphy and sampling sites (designated by letters, which have no stratigraphic significance) are shown in Fig. 2. Most sites yielded data that can be interpreted as either reversed or normal polarity magnetizations. Normal and reversed magnetizations occur at all three localities, in both diamicts and paleosols.

Examples of AF demagnetization of normal magnetization are shown in Fig. 3 (paleosol 1 at Mokowan Butte), Fig. 4 (paleosol 4 at Saint Mary Ridge), and Fig. 5 (thermal demagnetization of calcrete from unit 5 on Saint Mary Ridge). These examples are single-component magnetizations with good end points. There is excellent linear decay to the origin on the orthogonal plots in Fig. 5, and less regular decay in Figs. 3 and 4.

Examples of reversed magnetization are shown in Figs. 6 and 7. The calcrete of till 2 from Two Medicine Ridge shows linear decay to the origin and an excellent end point, signifying a single-component magnetization (Fig. 6). Sediment pockets in till 1 at Two Medicine Ridge yield poor orthogonal plots and variable directions (Fig. 7). In this example, there is a large Brunhes overprint, and the underlying magnetization is unmistakably reversed.

Two sites at Two Medicine Ridge (E, F) have magnetizations that are neither normal nor reversed. Site E has both positive and negative inclinations in different specimens, which changed little upon cleaning in fields of up to 175 mT (Fig. 8). Directions at site F also are widely scattered.

Thermal demagnetization curves for calcrete specimens show sharp decreases in the intensity of magnetization between 675 and 700°C, indicating that the magnetization is carried by hematite (Fig. 9). The slow decline between 200 and 600°C presumably reflects the presence of a range of hematite grain sizes. Directions of magnetization did not change during thermal demagnetization (Figs. 5, 6).

Fig. 1. Outcrops of Kennedy Drift in the study area. The Lewis Thrust marks the eastern limit of the Rocky Mountain fold and thrust belt.

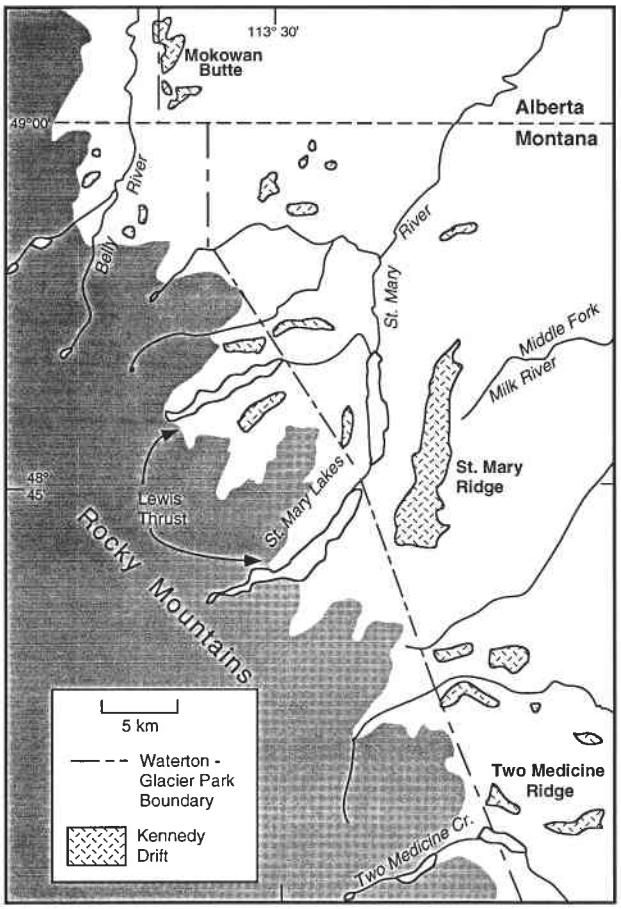


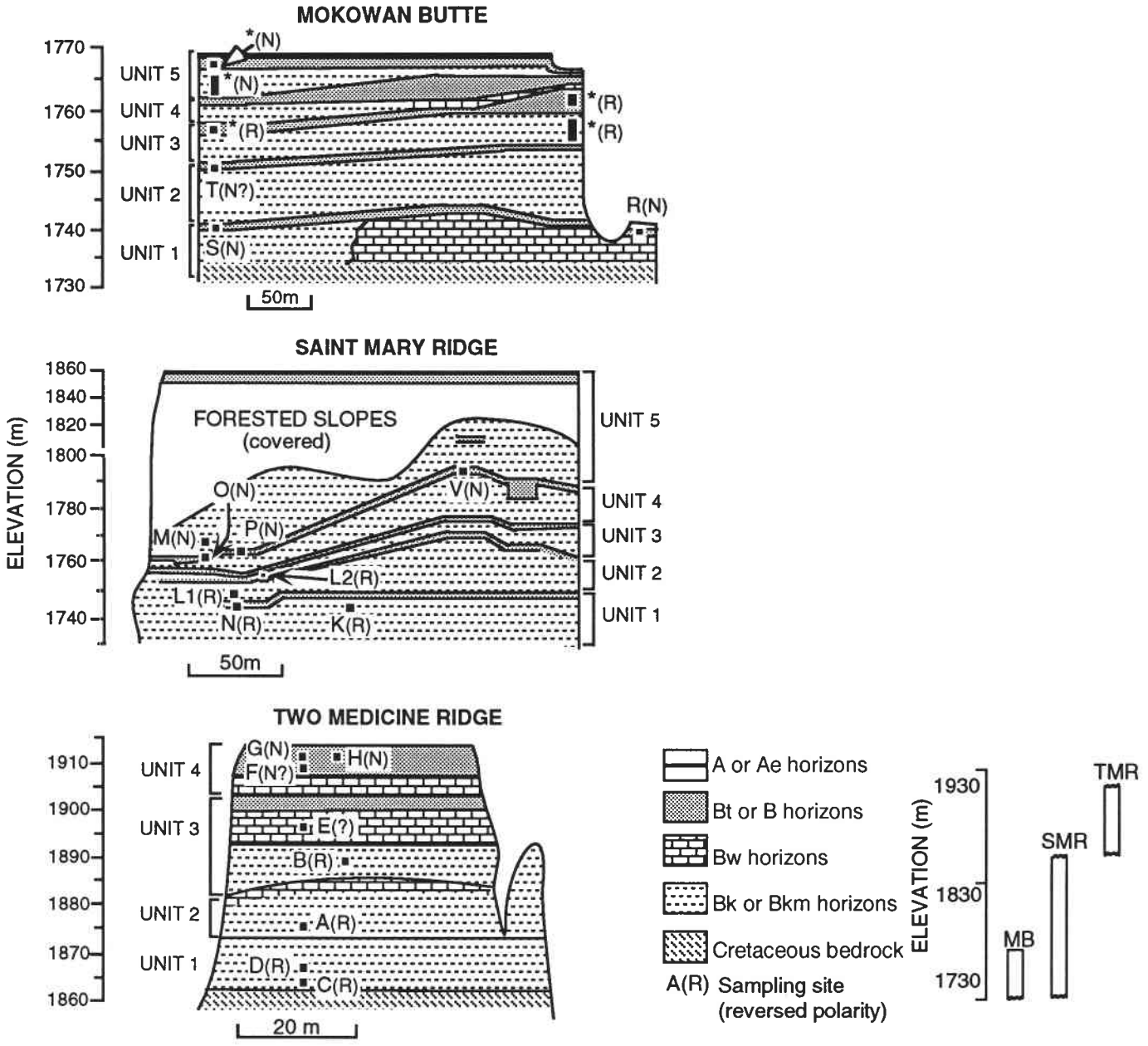
Table 1 lists site mean directions after the removal of Brunhes overprints and includes the data of Barendregt et al. (1991). Polarity is clearly defined in all cases, except sites E and F. Within-site scatter is highly variable, with Fisher's precision, k, ranging from 11 to 113.

Mokowan Butte

Karlstrom (1987) described the till–paleosol sequence at Mokowan Butte. The paleosol of unit 2 had not been identified previously at the major sampling site (a south-facing landslide scarp); however, we identified and sampled this paleosol (site T, Fig. 2). The lowermost diamicton (unit 1, site S) on the landslide scarp contains clasts that are more rounded and better sorted than those in overlying sediments, and may have been deposited in an alluvial or glacial outwash setting. However, the correlative diamicton (site R) on a nearby roadcut is clearly of glacial origin.

Previous studies of unit 2 yielded scattered data for which polarity could not be determined; unit 3 had well-defined reversed polarity, unit 4 gave rather scattered data but a

Fig. 2. Stratigraphy and sampling sites at Mokowan Butte, Saint Mary Ridge, and Two Medicine Ridge. Length of darkened bar indicates the thickness sampled; *, sampling sites of Barendregt et al. (1991). Elevations of the sections are shown at lower right. The major horizons of the soils are designated A, Ae, and B. A is the surface horizon, containing organic material; Ae is a near-surface horizon, from which material has been leached; and B is a subsurface horizon, which has been enriched by illuviation from the above horizons. Lower-cased letters indicate the nature of weathering: k, secondary calcium carbonate; m, continuous cementation; w, alteration due to leaching and (or) weathering; t, translocated clay (White 1979). Part or all of the Bkm horizons consists of calcrete. Lenses of fine sediment occur in the unindurated horizons (Bw or Bk) tills. Modified from Karlstrom (1987).



generally reversed polarity, and unit 5 had normal polarity (Barendregt et al. 1991; Table 1). In the present study, sampling was restricted to units 1–3.

We sampled the paleosol of unit 1 at sites R and S (Fig. 2). At site S, the normal magnetization of Fig. 3 was observed. About 30% of the intensity was retained after AF treatment in 80 mT, and directions remained consistent until 100 mT and were systematically different from the present Earth's field (Table 1). Site R has more scattered directions, but they are generally similar to those from site S (Table 1). Because of the high coercivities seen in the samples, we suggest that the magnetization is original and not a Brunhes-age overprint.

The paleosol of unit 2 was sampled at site T. The natural remanent magnetization (NRM) directions are widely scattered, and partial demagnetization does not decrease the scatter significantly. Polarity is normal, but the site mean direction does not differ significantly from the present Earth's

Fig. 3. AF demagnetization (steps of 10 mT up to 100 mT) of paleoargillic horizon, unit 1 (site S), at Mokowan Butte. Top, directional changes. Filled circles indicate downward inclinations. Bottom, orthogonal plots (normal polarity).

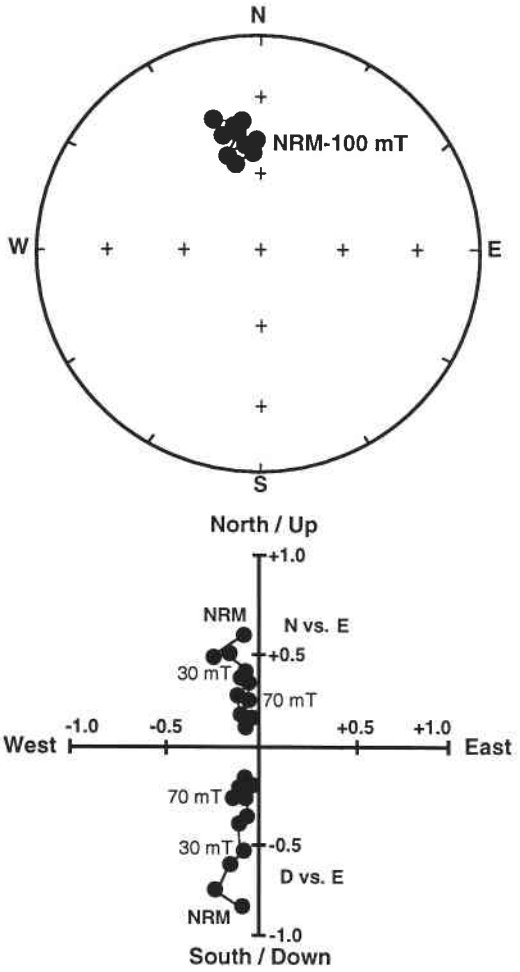
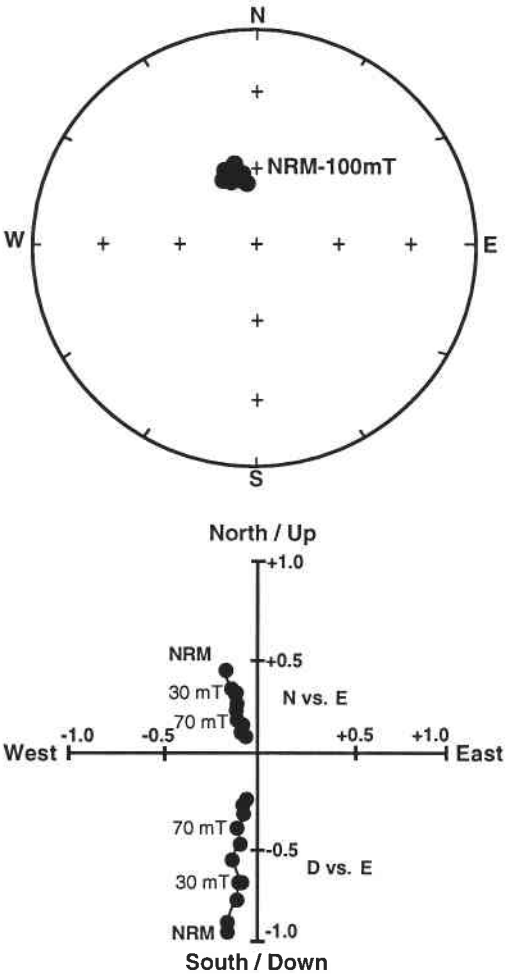


Fig. 4. AF demagnetization of paleoargillic horizon, unit 4 (site V), at Saint Mary Ridge. Symbols as in Fig. 3. Normal polarity.



field. The intensity decays more rapidly than that of unit 1 (Fig. 10). Thus, the magnetization may be entirely a Brunhes overprint.

The initial NRM directions we observed from unit 3 were widely scattered. The specimens were thermally demagnetized after being soaked in a solution of water glass, in an attempt to keep them from falling apart at high temperatures. The results are similar to those of Barendregt et al. (1991), confirming reversed polarity for unit 3.

In summary, unit 1 has normal polarity, unit 2 has normal polarity but the magnetization could be a Brunhes overprint, units 3 and 4 are reversed, and unit 5 has normal polarity.

Saint Mary Ridge

There are at least five tills and four paleosols exposed on a landslide scarp at Saint Mary Ridge (Karlstrom 1981, 1988). Older till–paleosol units may exist beneath unit 1. Two of the five paleoargillic horizons (units 1, 4) and calcrete-horizons from units 1, 2, 3, and 5 were sampled (Fig. 2). The paleoargillic horizons are generally thinner and less developed than those at Mokowan Butte and Two Medicine Ridge, and consist of yellowish-red, leached, argillic horizons, and (or) reddish-brown, leached, Bw horizons overlying 4–15 m thick calcretes (Karlstrom 1988). Calcretes are more common here than at Mokowan Butte.

Specimens from units 1 and 2 (sites K, L1, N) yielded reversed site mean directions with low precision (Table 1). The decay in intensity during demagnetization was not smooth and directions changed erratically, although they are clearly reversed (Fig. 7).

Unit 3 (site L2) is also reversed. Its precision is higher than that of the three sites of units 1 and 2. Its mean direction also differs significantly from the directions of units 1 and 2, although similar behaviour is exhibited during demagnetization.

The paleoargillic horizon of unit 4 (sites O, P, V) has normal polarity. At site P, the magnetization has a shallow inclination, but the paleosol at this site is not horizontal, and may have slumped, affecting the direction. All samples exhibit similar behaviour during demagnetization (Fig. 4).

Unit 5, sampled at site M, has single-component magnetization of normal polarity (Fig. 5).

In summary, at Saint Mary Ridge reversed polarity beds (units 1, 2, 3 underlie normal polarity beds (units 4, 5). Generally, sites with normal polarity have high precision and

Fig. 5. Thermal demagnetization of calcrete in uppermost till, unit 5 (site M), Saint Mary Ridge. Symbols as in Fig. 3. Normal polarity.

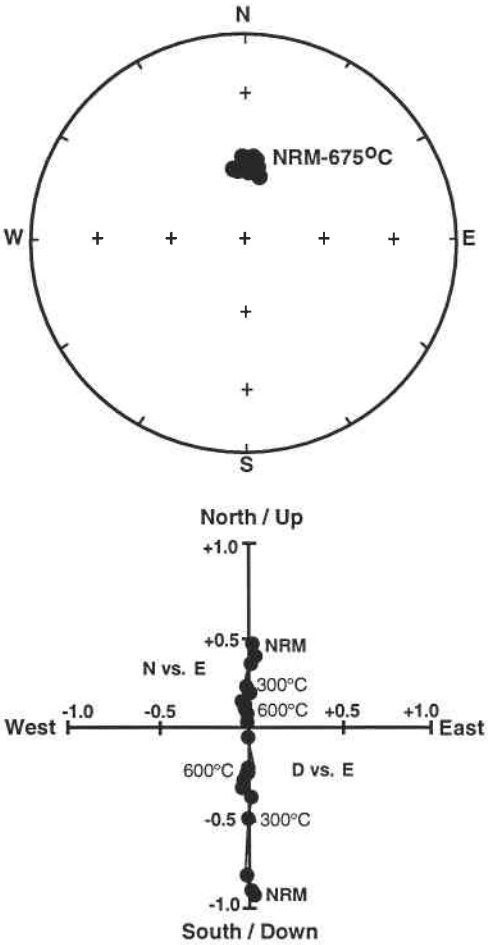
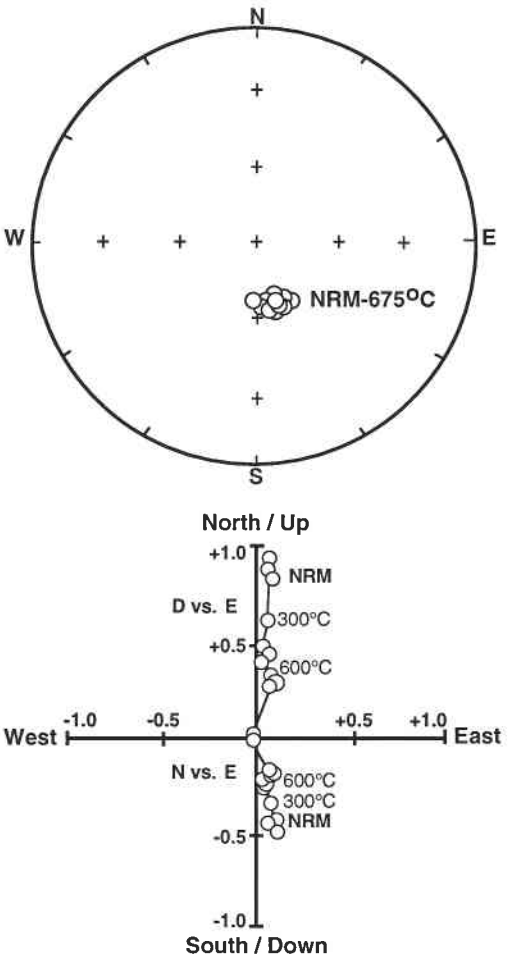


Fig. 6. Thermal demagnetization of calcrete in till, unit 2 (site A), Two Medicine Ridge. Symbols as in Fig. 3, except that open circles in the stereogram indicate upward inclinations. Reversed polarity.



the decay of intensity is smooth and linear to the origin on orthogonal plots, whereas reversed sites have lower precision and the decay of intensity is less smooth. This may be due to the greater frequency of pebbles or to the presence of larger, less easily removed Brunhes-age overprints in older sediments.

Two Medicine Ridge

Karlstrom (1981, 1988) stated that up to three buried paleosols and four tills are present on the retreating cliff face of Two Medicine Ridge. Richmond (1986) reported samples with reversed polarity; however, their position within the sequence is uncertain. In the present study, samples were collected from eight sites in units 1–4.

The oldest till has no associated paleosol, but contains pockets of fine-grained sediment sampled at sites C and D. The demagnetization behaviour is not ideal, but reversed polarity is apparent at both sites (Table 1; Fig. 7).

Calcrete horizons in till units 2 and 3 (sites A, B) are reversed. There is a smooth decay in intensity during demagnetization and the magnetization direction does not change (Fig. 6). Precisions are higher than in till unit 1 (Table 1). A pocket of fine-grained sediment within a leached Bw horizon in till unit 3 (site E) was also sampled. Directions are scattered and the polarity is indeterminate (Fig. 8). The magnetizations, however, are very stable, and directions do not change in fields of up to 175 mT. The deposition of diagenetic hematite during weathering and soil development in the Brunhes polarity chron could have produced a normal magnetization of high magnetic stability overprinted on an earlier reversed magnetization. Alternatively, if the sediments were deposited very quickly, they could record a transition from reversed to normal polarity. We regard the former as the more likely explanation.

The paleosol of unit 4 was sampled at three sites: site F, near the bottom, and sites G and H, towards the top. Specimen directions at site F are scattered about a vertical downwards direction. This could be a transitional direction during reversal of the geomagnetic field, but the error is large (17°) and we consider that the direction falls within the range expected for secular variation. Sites G and H exhibit well-defined normal polarity (Fig. 5).

In summary, at Two Medicine Ridge magnetically reversed units 1 and 2 and lower unit 3 are overlain by normal polarity unit 4. The upper site in unit 3 has an oblique direction and we interpret this to be probably caused by variable partial overprinting.

Fig. 7. AF demagnetization of fine-grained sediment within till, unit 1 (site C), Two Medicine Ridge. Symbols as in Fig. 3, except that open circles in the stereogram indicate upward inclinations. Reversed polarity.

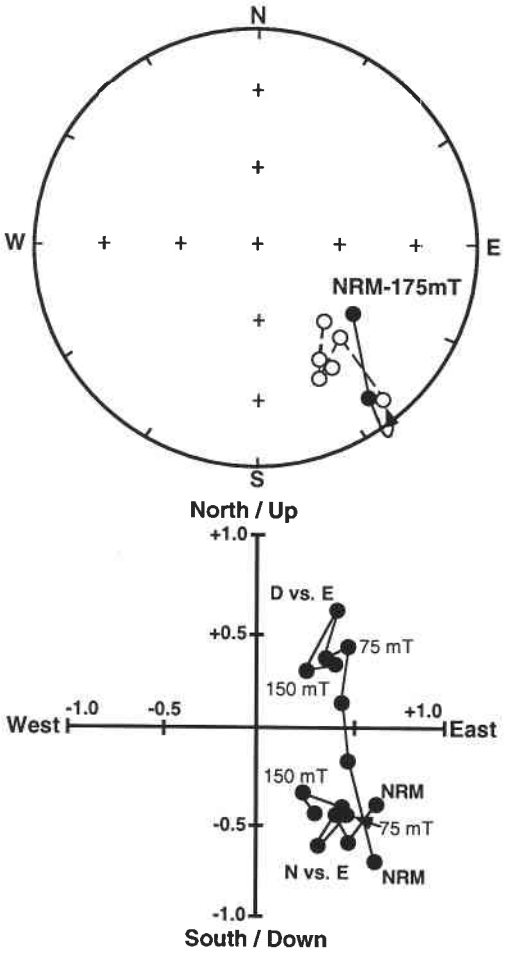


Fig. 8. Magnetization direction in till, unit 3 (site E), Two Medicine Ridge. After AF demagnetization at 80 mT, individual directions change, but the general distribution changes little. Symbols as in Fig. 3, except that open circles indicate upward inclinations.

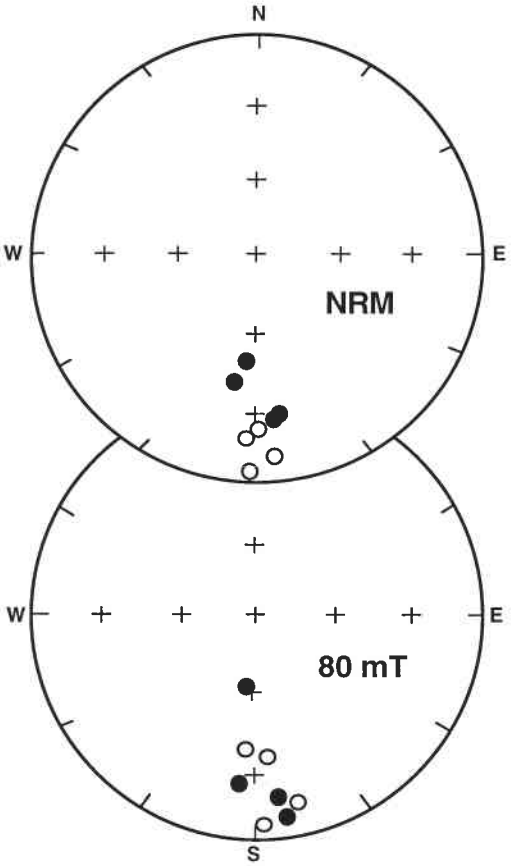
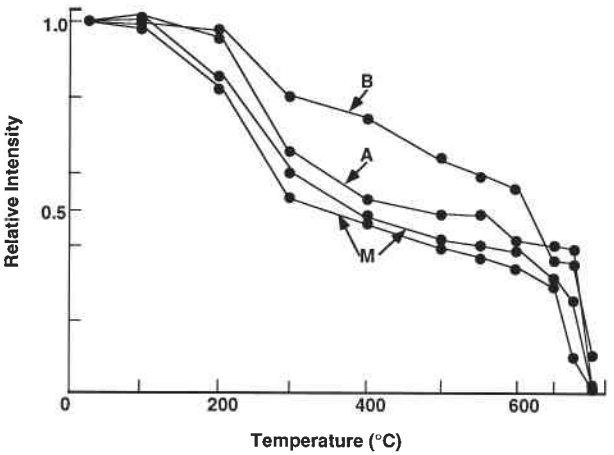


Fig. 9. Thermal demagnetization curves for calcrete specimens from sites A and B at Two Medicine Ridge, and site M at Saint Mary Ridge.



Discussion

Origin and duration of magnetization records

The presence of hematite in the paleosols and calcretes is indicated by their reddish-brown colour and by their unblocking temperatures (Fig. 9). The precision, k, is, in general, inversely proportional to the proportion of pebbles present, and it is likely, therefore, that the remanent magnetization is carried by the matrix, in which hematite was precipitated from groundwater. Hence, it is probable that the magnetization is a chemical remanent magnetization acquired as the hematite grains grew and exceeded the critical blocking diameter (1 μm for hematite).

Schwertmann (1985) has suggested that hematite forms at mean annual temperatures above 7°C (present mean annual temperatures in the sampling area range from 2 to 5°C). Thus, the magnetization-carrying hematite (in both the paleosols and the tills) probably formed at the same time as the paleosols, that is, during warm interglacial intervals. Karlstrom (1987, 1988) suggested that up to several hundred thousand years would be required to form each paleosol. If so, the hematite present in a particular paleosol is likely to have attained the critical blocking diameter and acquired its magnetization over a comparable period, that is, over a polarity chron rather than a subchron. This is referred to as the *integrated record* hypothesis. Alternatively, hematite of the appropriate blocking diameter may have been formed over a comparatively short interval within the much longer period that the paleosol developed, yielding only a short

Table 1. Summary of magnetizations.

Site	Unit	Polarity	n	D (°)	I (°)	k	α_{95}
Two Medicine Ridge							
H	4	Normal	5/5	352	60	60	11
G	4	Normal	5/5	14	60	64	10
F	4	Normal	10/10	225	84	10	17
E	3	Transitional?	8/8	176	1	6	24
B	3, B_{km}	Reversed	16/17	193	−74	58	5
A	2, B_{km}	Reversed	13/13	167	−63	113	4
D	1	Reversed	13/14	166	−45	9	15
C	1	Reversed	9/9	166	−40	19	12
Saint Mary Ridge							
M	5, B_{km}	Normal	10/10	6	61	483	2
V	4	Normal	12/12	360	65	152	4
P	4	Normal	4/5	309	37	74	11
O	4	Normal	4/4	344	74	84	10
L2	3, B_{km}	Reversed	10/10	207	−62	29	9
L1	2, B_{km}	Reversed	7/7	162	−53	16	16
N	1	Reversed	12/13	156	−53	16	11
K	1, B_{km}	Reversed	9/14	154	−59	11	16
Mokowan Butte							
*	5	Normal	4	348	69	55	13
*	5	Normal	21	356	74	19	7
*	3	Reversed	24	151	−46	6	13
*	3	Reversed	34	149	−57	10	13
T	2	Normal	11/11	63	55	3	40
S	1	Normal	5/5	357	58	56	10
R	1	Normal	6/7	354	46	10	23

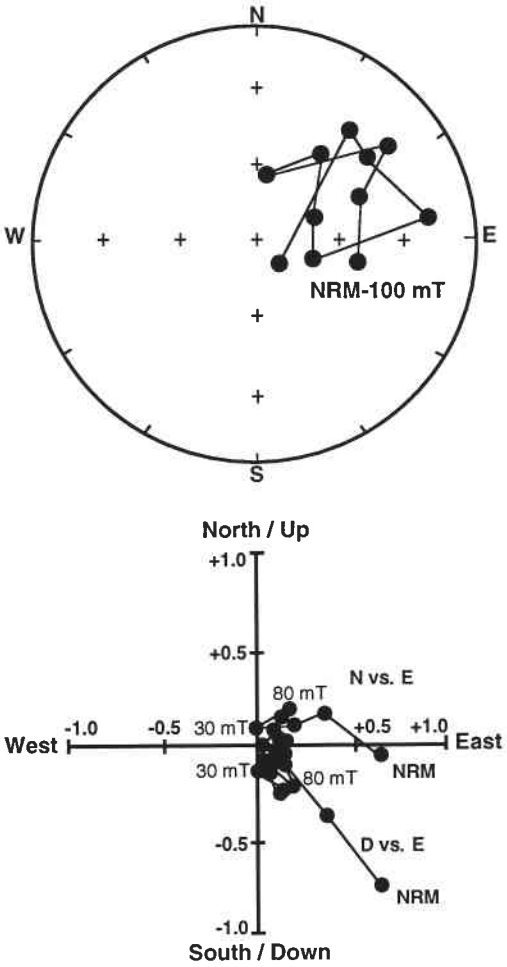
Notes: Sites are grouped by locality. * in the Site column, data of Barendregt et al. (1991); B_{km} in the Unit column, calcretes; 4/5, e.g., in the n column, 4 specimens out of a total of 5 used to determine the site mean; D, declination; I, inclination; k, precision parameter, α_{95}, circle of confidence ($P = 0.05$). Several B_{km} calcrete samples were thermally demagnetized at temperatures up to 680°C (Fig. 9). The sediment samples and the remainder of the B_{km} samples were demagnetized in alternating fields of up to 180 mT.

magnetic record. This is referred to as the *spot record* hypothesis. The wide range of soil development present in the Kennedy Drift suggests a third hypothesis—the *integrated spot record hypothesis*. According to this hypothesis, well-developed paleosols would have magnetic records that integrate the field over long intervals, whereas less well-developed paleosols would have a shorter record.

Integrated record hypothesis

Figure 11 shows the observed polarity sequences interpreted in light of the integrated record hypothesis. A zone of normal polarity occurs at the top of each section. The normal deposits include the surface soils (Barendregt et al. 1991), suggesting that these are Brunhes magnetizations. At Saint Mary Ridge, a uranium-trend age of 440 ± 120 ka on normally magnetized unit 5 indicates that early Brunhes magnetizations are present. Furthermore, if, as Karlstrom (1987, 1988) has argued, soil development occurred over $>10^5$ years, then the upper normally magnetized paleosols and

Fig. 10. AF demagnetization of paleoargillic horizon, unit 2 (site T), Mokowan Butte. Symbols as in Fig. 3. Possible Brunhes overprint (compare with Fig. 3).



diamict would span a substantial portion of the Brunhes Chron, as depicted in Fig. 11.

At Two Medicine Ridge, sediments of uncertain polarity (till−paleosol unit 3) occur between sediments of normal polarity and underlying sediments of reversed polarity. Although Brunhes-age overprinting is the most likely explanation, the possibility that a record of the Matuyama−Brunhes transition is present cannot be excluded.

All three localities record a zone of reversed polarity below the upper normal zone. As this reversed zone directly underlies the normal polarity zone at Saint Mary Ridge and Mokowan Butte, it is reasonable to postulate that the reversely magnetized sediments are of Matuyama age (2600−780 ka). Alternatively, they could have been formed during the Gilbert Reversed Chron (5200−3400 ka). Boelstorff (1978) found evidence for glaciations in the mid-continental U.S.A. as old as 2200 and 2400 ka. Thus, although a Gilbert age for the reversed sediments cannot be entirely ruled out, a Matuyama age is much more likely.

At Mokowan Butte, sediments of normal polarity occur below the reversed zone. The magnetization of unit 1 is stable (Fig. 3), and could be a record of the paleofield during the Gauss Normal Polarity Chron (3400−2600 ka). This

indicates a minimum age of 2600 ka for the oldest till–paleosol unit at this locality.

Spot record hypothesis

According to this hypothesis, the magnetization at each sampling site records only a brief period of time. The interpretation that the upper normal and underlying reversed sediments at the three localities formed during the Brunhes and Matuyama chrons remains the same, but the magnetization of the oldest till–paleosol unit at Mokowan Butte could have formed during the Olduvai Subchron (1950–1770 ka) within the Matuyama Reversed Chron. This would imply a minimum age of 1770 ka for this unit. The reversed paleosols and tills would have formed during the interval 1770–780 ka, and the lower normal paleosol during the 180 ka interval of the Olduvai Subchron. This interpretation provides ample time for the deposition of the reversed sediments; however, it is unlikely, although not impossible, that one normal paleosol and the intervening till would have formed in that 180 ka interval.

Alternatively, the normal unit(s) at Mokowan Butte could have been deposited and magnetized during the Jaramillo Subchron (1070–990 ka). This possibility is rejected for two reasons. First, the Jaramillo event is almost certainly too short (80 ka) for the formation of two paleosols in two successive tills, each representing a separate glaciation. Second, it is unlikely that the extremely thick and highly weathered paleosols in units 3 and 4 at Mokowan Butte could have developed in the short interval (990–780 ka) between the end of the Jaramillo Subchron and the beginning of the Brunhes Chron.

Integrated spot record hypothesis

According to this hypothesis, the magnetizations of the well-developed paleosols in units 3, 4, and 5 at Mokowan Butte, units 3 and 4 at Two Medicine Ridge, and units 4 and 5 at Saint Mary Ridge integrate the geomagnetic field over periods of 10^5 years, whereas the magnetizations of the less-developed paleosols in units 1 and 2 at Mokowan Butte, units 1, 2, and 3 at Saint Mary Ridge, and unit 2 at Two Medicine Ridge provide spot readings of the field.

The interpretation of the upper normal and underlying reversed units (Brunhes and Matuyama chrons, respectively) remains the same. The well-developed paleosols in the two reversed units (3 and 4) at Mokowan Butte indicate a lengthy interval of soil formation and magnetization, rendering a Jaramillo age for the lowermost normal units unlikely. While till deposition and paleosol formation during the Olduvai Subchron cannot be ruled out, it is more likely that it occurred in the Gauss Chron for the reasons just given.

Preferred interpretation

The main uncertainty in the magnetostratigraphic assignments is whether normally magnetized unit 1 (and possibly unit 2) at Mokowan Butte formed during the Gauss Normal Polarity Chron, the Olduvai Subchron, or the Jaramillo Subchron (Fig. 11). The above discussion, while by no means conclusive, favours the first possibility, that is, that the lower normal paleosol(s) at Mokowan Butte were formed during the Gauss Chron, the reversed paleosols and tills during the Matuyama Chron, and the upper normal paleosols and tills during the Brunhes Chron. This interpretation, based as it is on the integrated record hypothesis, is dependent on the assumption, detailed above, regarding the length of time recorded by magnetization.

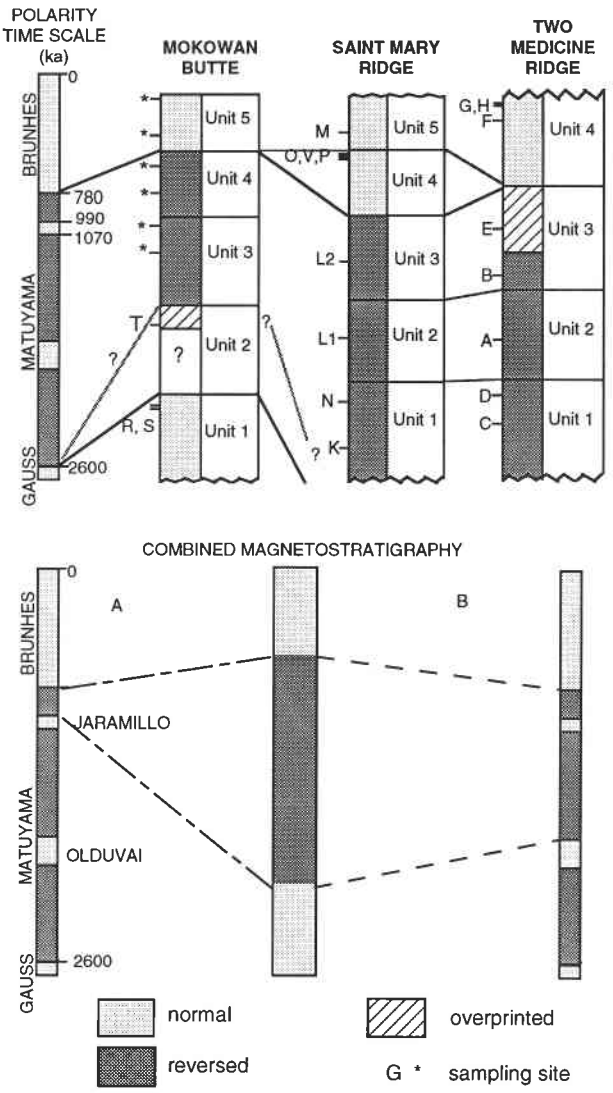


Fig. 11. Top, correlation and assigned ages of the three localities, based on our preferred interpretation, which assumes till–paleosol units 1, and possibly 2, at Mokowan Butte are Gauss. Bottom, alternative interpretations (see text). Stratigraphic thicknesses in Fig. 2. The time scale is that of Shackleton et al. (1990). *, sites of Barendregt et al. (1991).

Correlation among sections

Unit 5 at Mokowan Butte, unit 5 at Saint Mary Ridge, and unit 4 at Two Medicine Ridge, the uppermost units at each locality, are all of normal polarity, and are likely correlative (Fig. 11). Unit 4 at Saint Mary Ridge, which is also of normal polarity, has no obvious counterpart at other localities. Hence it appears that two Brunhes-age tills occur only at Saint Mary Ridge. We suggest that reversed unit 3 at Two Medicine Ridge correlates with the uppermost reversed units at Mokowan Butte (unit 4) and Saint Mary Ridge (unit 3).

Units 1 and 2 at both Saint Mary Ridge and Two Medicine Ridge also are reversed and may correlate. Unit 3 at Mokowan Butte is reversed and could correlate with either or both of the reversed units at Saint Mary Ridge and Two Medicine Ridge. Mokowan Butte has at least one normal polarity unit (unit 1) below the reversed units. The normal polarity of unit 2 at Mokowan Butte may be a Brunhes age overprint, obscuring an originally reversed polarity. No normal polarity units were seen beneath the reversed polarity units at Saint Mary Ridge and Two Medicine Ridge.

Based on these correlations (Fig. 11), at least six glacial and six interglacial episodes are recorded in the Kennedy Drift. If the magnetization of unit 2 at Mokowan Butte is original, and not a Brunhes overprint, then seven glacial episodes are recorded. Additional older tills and paleosols may lie beneath the Saint Mary Ridge section, because the base of the section is not exposed.

Conclusions

Paleomagnetic study of tills and intercalated paleosols of the Kennedy Drift in Montana and Alberta shows that the sequence at Mokowan Butte consists of at least one normal polarity till–paleosol unit overlain by two reversed polarity units, which are in turn overlain by another normal polarity unit; the sequence at Saint Mary Ridge consists of three reversed polarity units overlain by two normal polarity units; and the sequence at Two Medicine Ridge consists of three reversed polarity units overlain by a single normal polarity unit. Magnetostratigraphic correlation of the till–paleosol units at the three localities shows that a minimum of six glacial and six interglacial episodes are recorded by the Kennedy Drift.

We interpret the normal polarity units at the top of all three sequences as having formed during the Brunhes Normal Chron, the underlying reversed units as having formed during the Matuyama Reversed Chron, and the normal unit(s) at the bottom of the Mokowan Butte sequence as having formed during the Gauss Normal Chron. These conclusions imply a minimum age of 2600 ka for the oldest unit and probable glaciation in this area.

Acknowledgments

We thank Dr. John Clague of the Geological Survey of Canada for a very careful and thorough review of this paper, and Richard Meyers for able assistance in the field. This research was supported by a Natural Sciences and Engineering Research Council of Canada grant to E. Irving and to R.W. Barendregt.

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